Stochastic interacting particles

- N “particles”

- Particle $i \in \{1, \ldots, N\}$, has state $Z_t^i \in E$ at time t

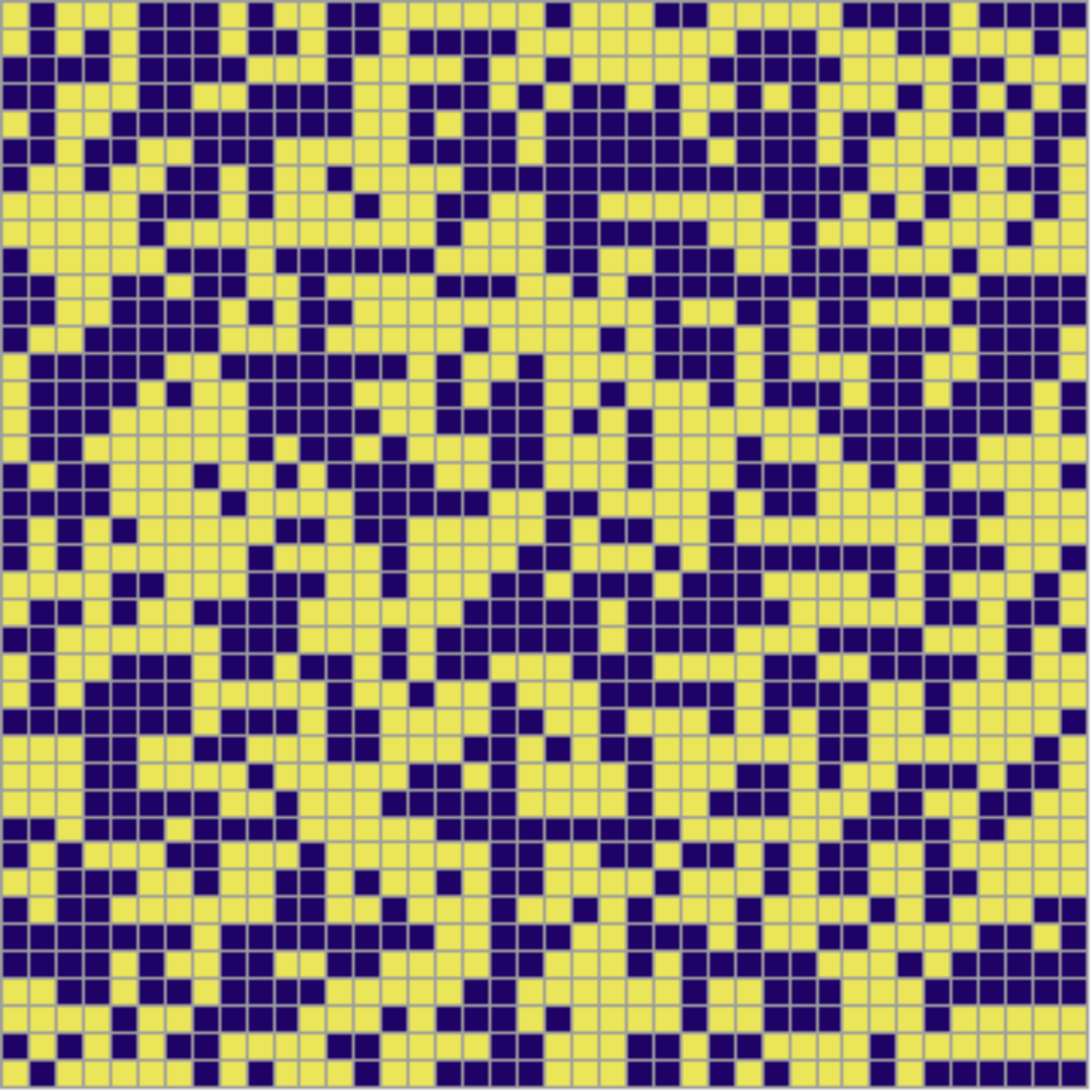
- Interacting: $Z^i_{t+\Delta t}$ depends on $\{Z^i_t\}_{i=1}^N$

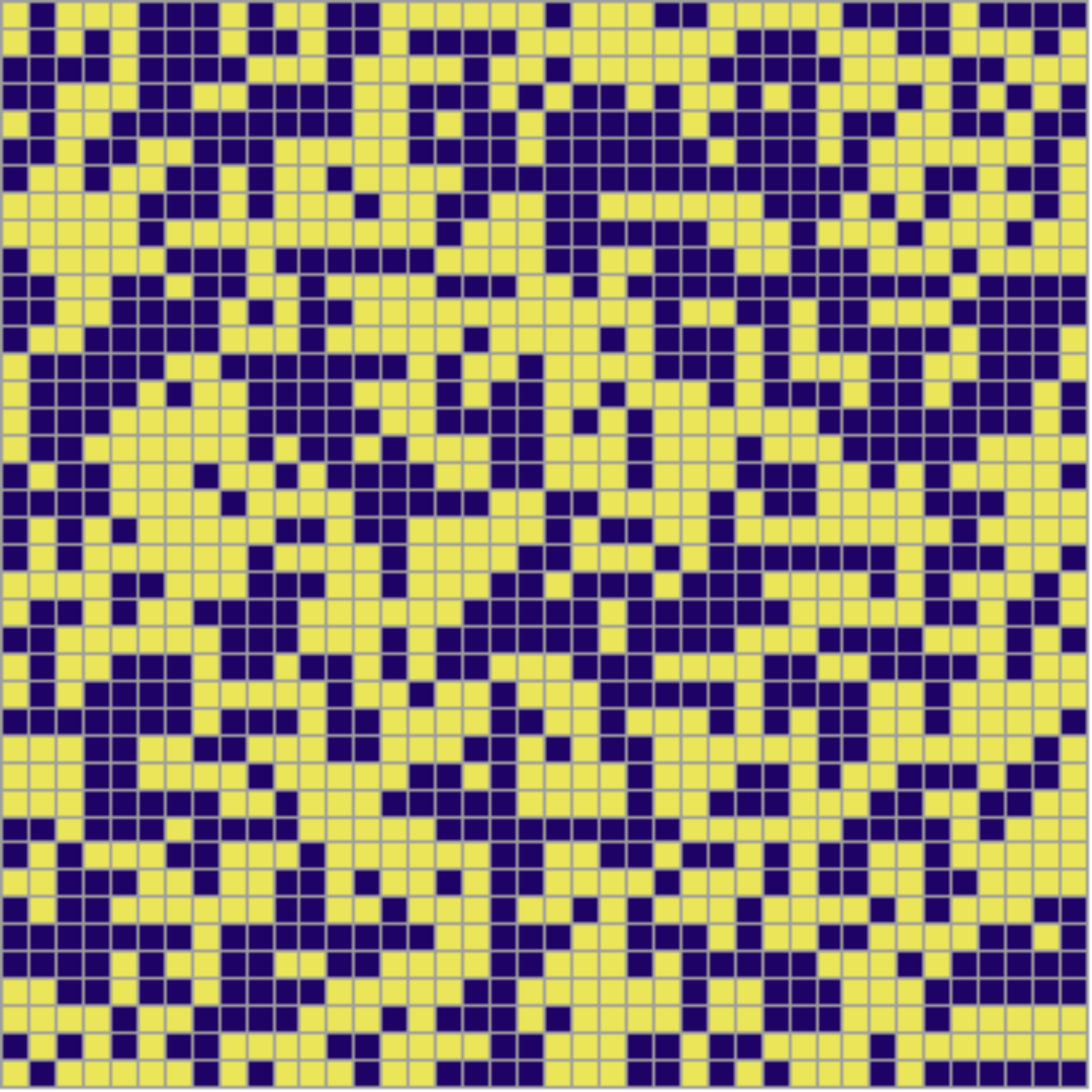
- Examples

- Ising model: $E = \{\pm 1\}$, discrete time

- Interacting stochastic processes? $\mathrm{d}X_t = b(X_t)\mathrm{d}t + \sigma(X_t)\mathrm{d}W_t$

- McKean-Vlasov models: $E = \mathbb{R}^d$, $\mathbb{T}^d$ or $\Omega \subset \mathbb{R}^d$





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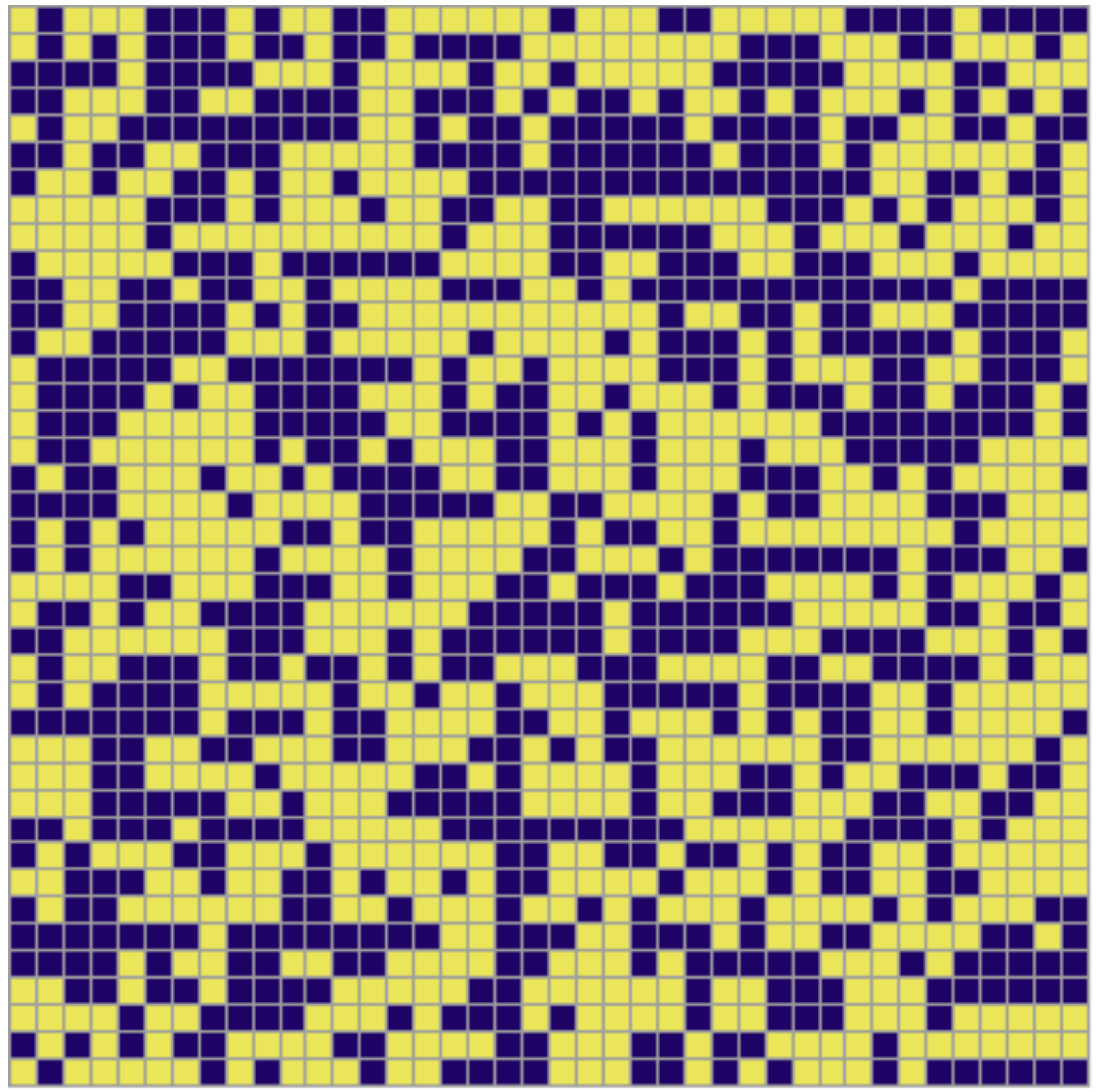

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McKean-Vlasov model

- Simple model: consider
  $\mathrm{d}X_t^i = \sum_{j=1}^N a(X_t^i - X_t^j)\mathrm{d}t + \sigma \mathrm{d}W_t^i$

- What is the behaviour depending on a, σ, N?